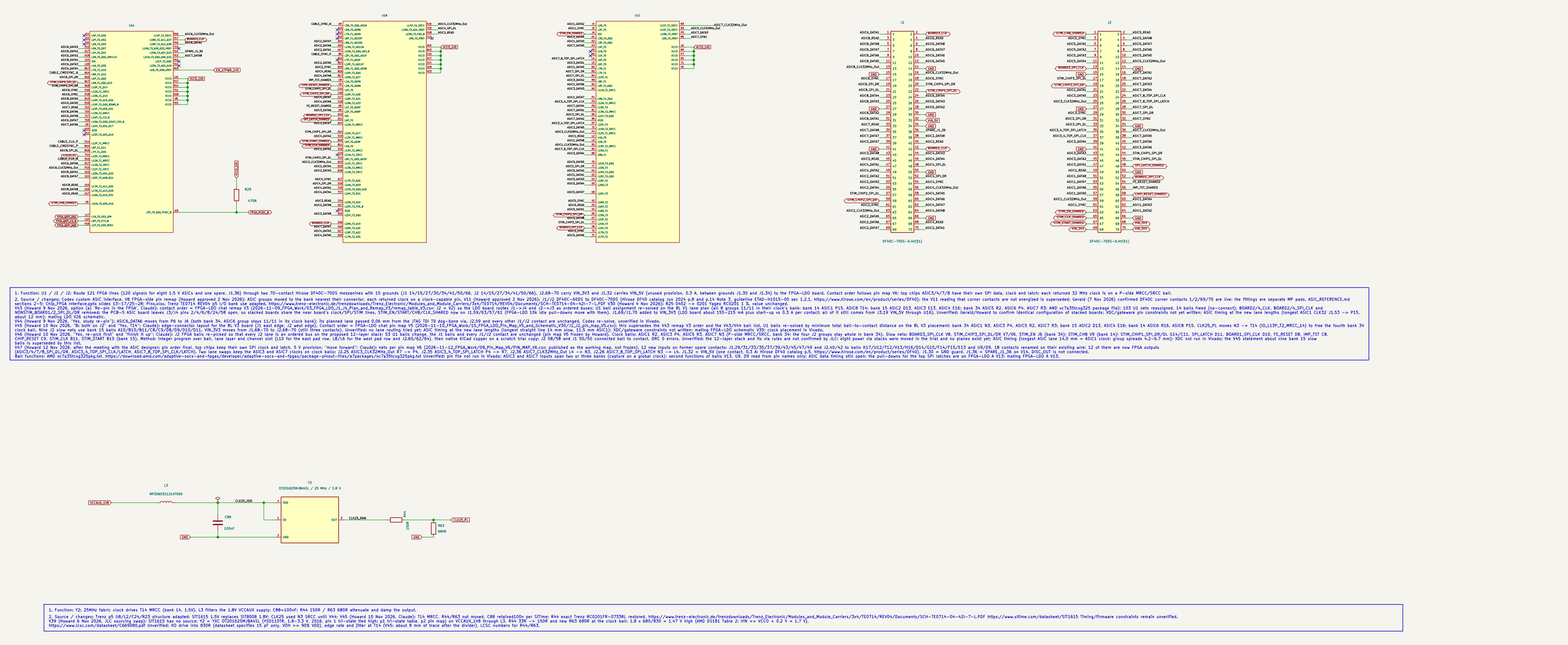
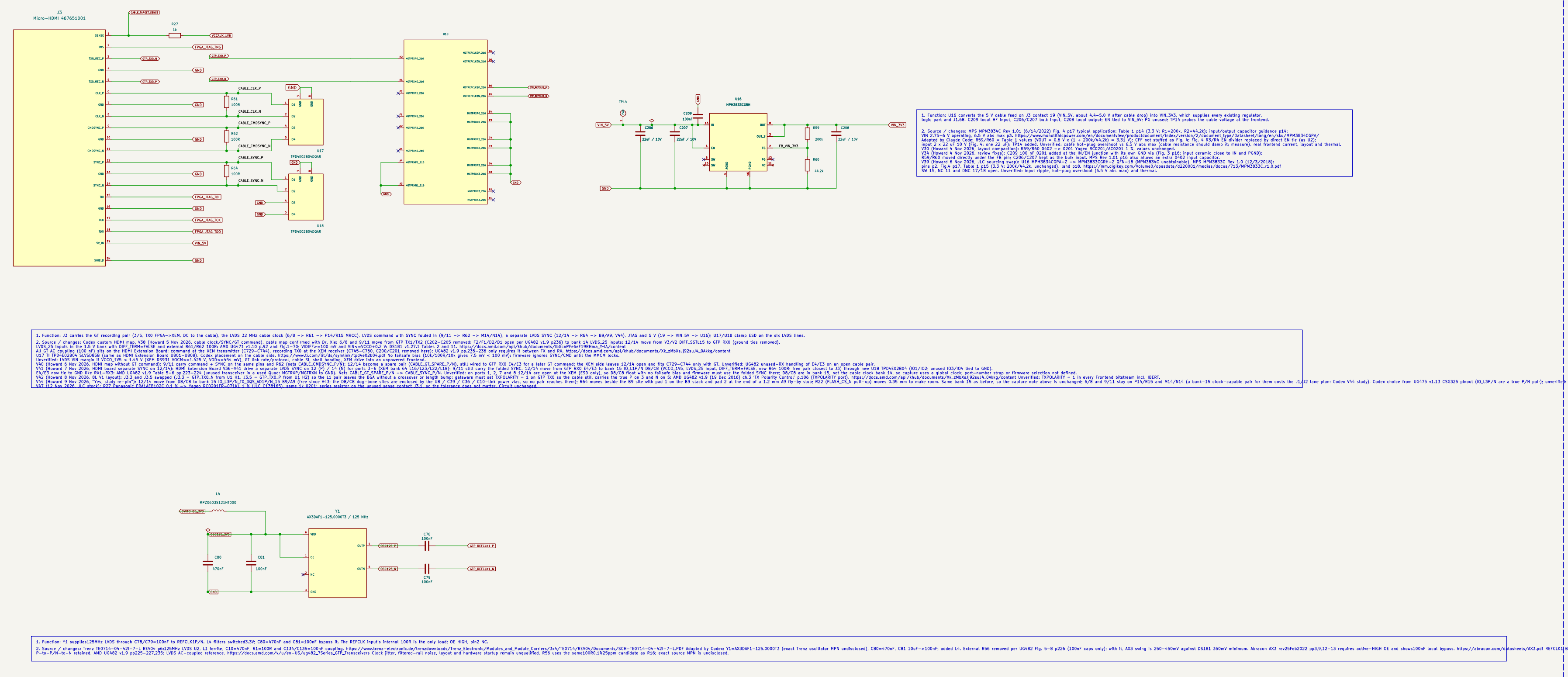
01. FPGA I/O + ASIC INTERFACES + 25 MHz CLOCK | Trenz p5

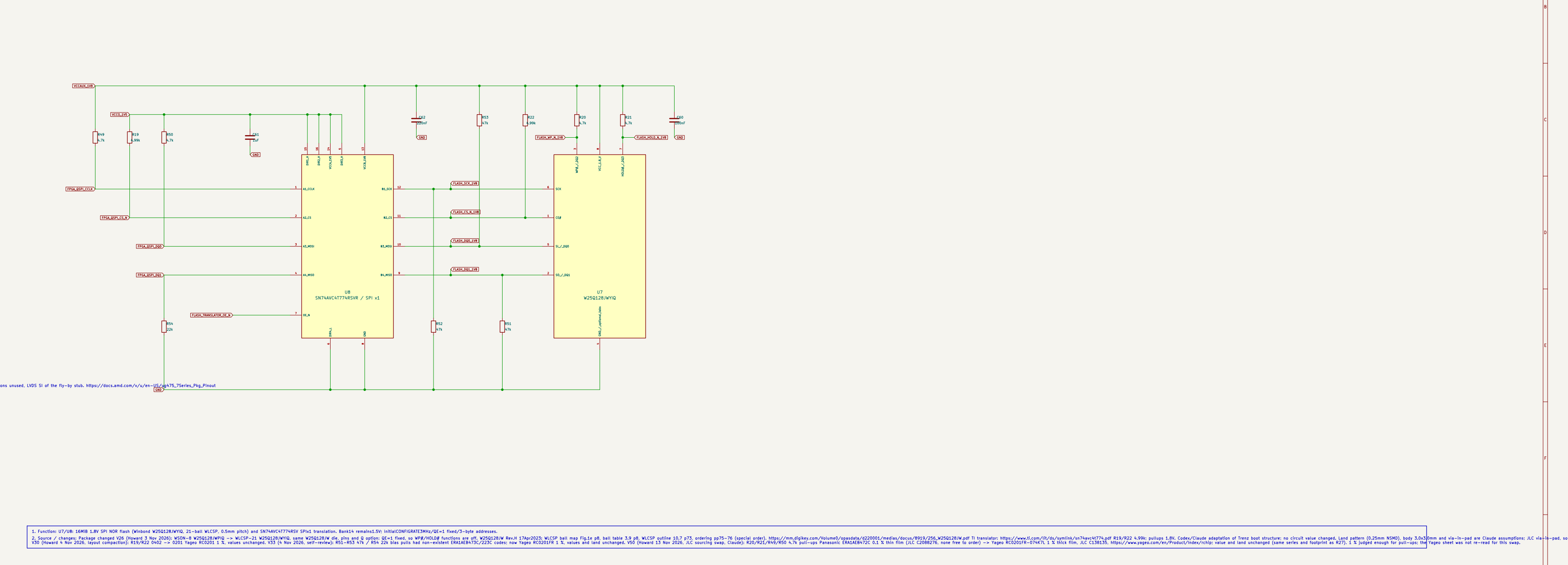


1. Function: U1 / J1 / J2: Route 121 FPGA lines (120 signals for eight 1.5 V ASICs and one spare, J1.36) through two 70-contact Hirose DF40C-70DS mezzanines with 15 grounds (J1 14/15/27/30/34/41/50/66, J2 14/15/27/34/41/50/66). J2.68-70 carry VIN_3V3 and J1.32 carries VIN_5V (unused provision, 0.3 A, between grounds J1.30 and J1.34) to the FPGA-LDO board. Contact order follows pin map V6: top chips ASIC3/4/7/8 have their own SPI data, clock and latch; each returned 32 MHz clock is on a P-side MRCC/SRCC ball.

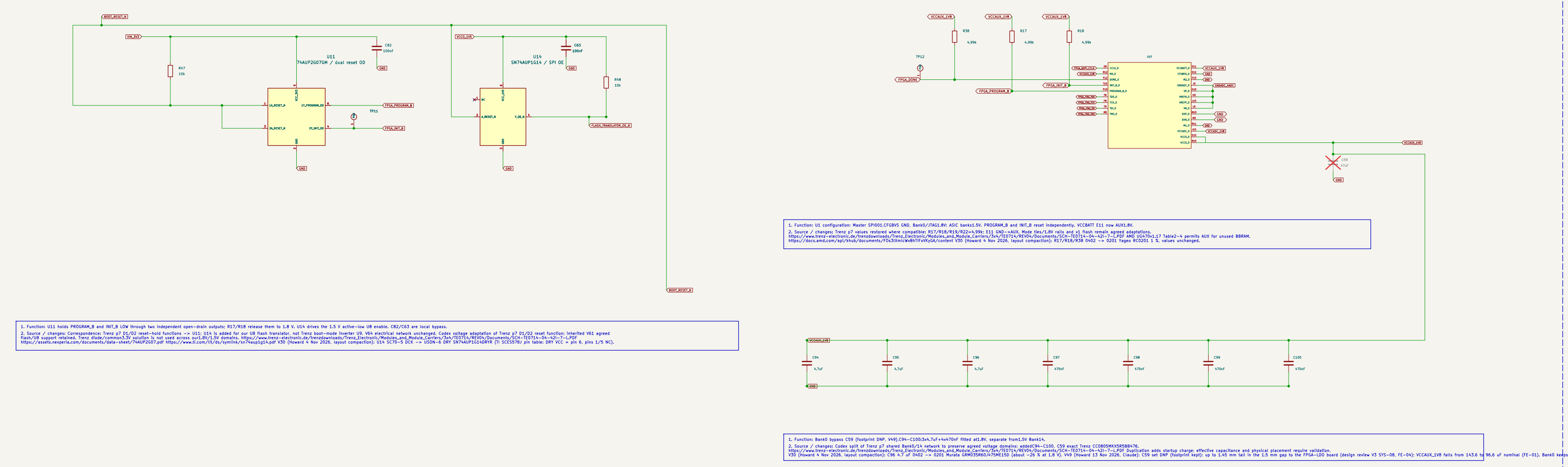
02. GTP CHANNELS + 125 MHz REFERENCE | Trenz p6



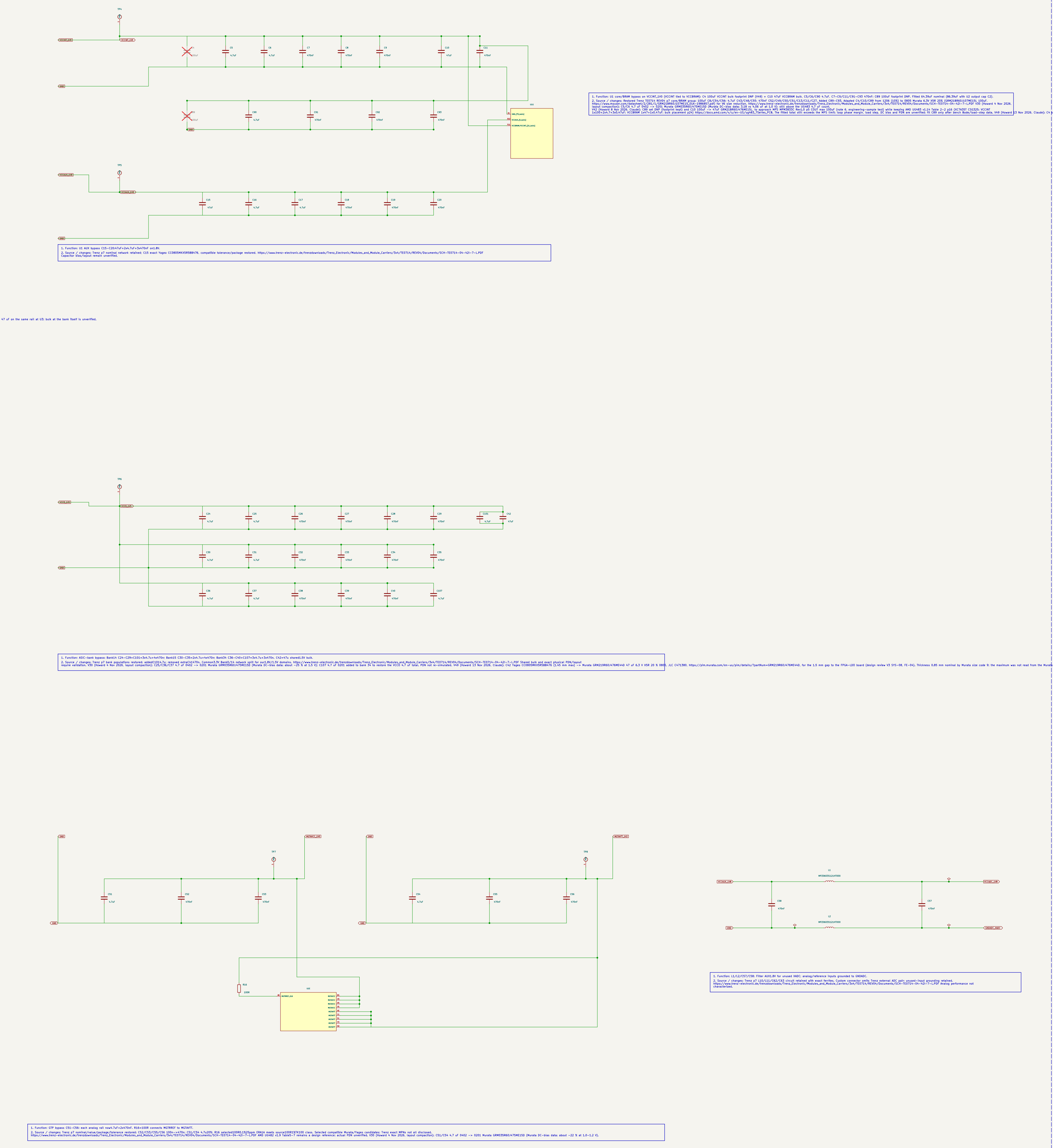
03. FLASH + SPI TRANSLATION | Trenz p6 adaptation



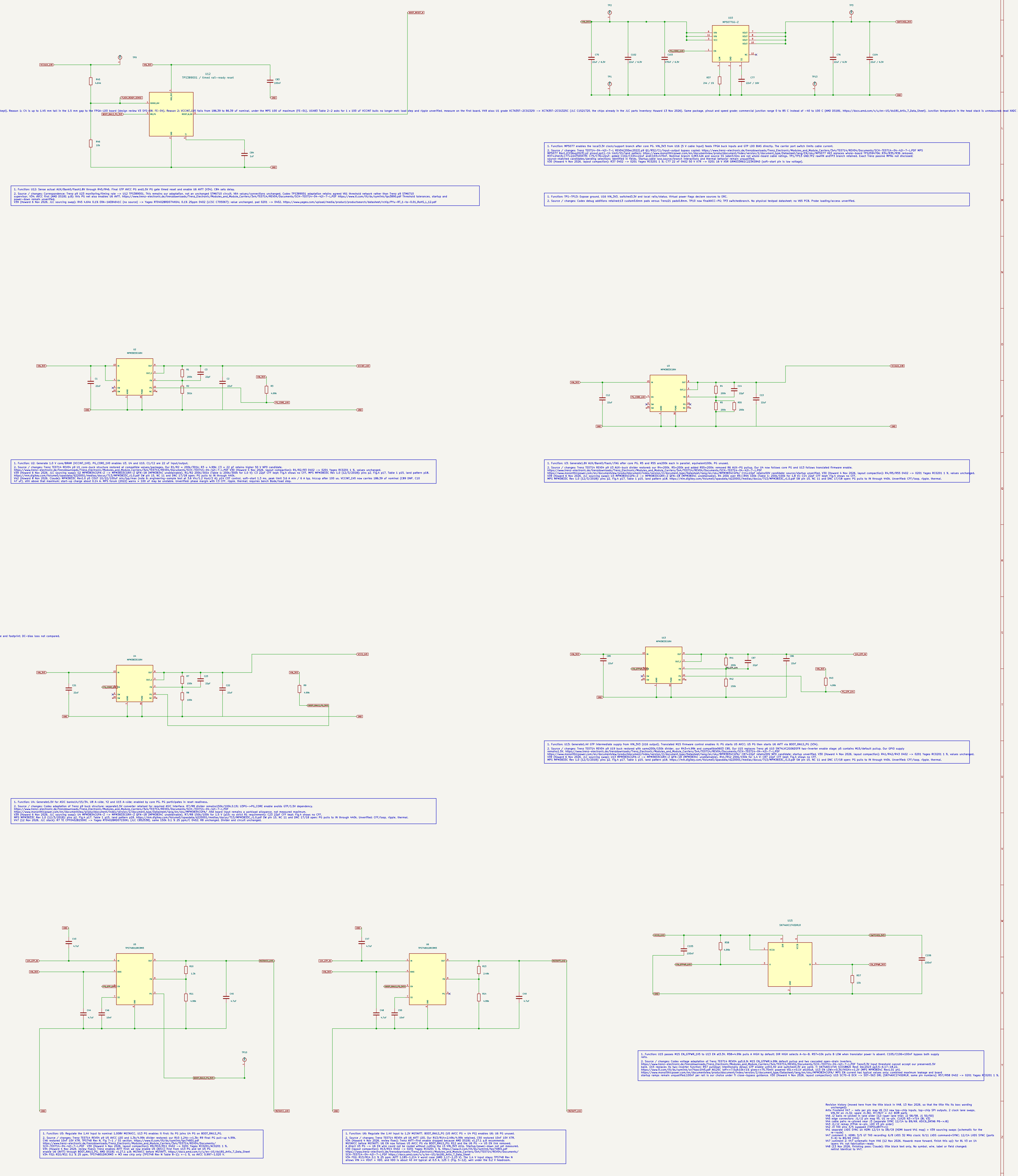
04. CONFIGURATION + RESET OUTPUTS | Trenz p7 roles



05. FPGA SUPPLY PINS + BYPASS + XADC | Trenz p7



06. REGULATORS + READINESS SUPERVISOR | Trenz p9 roles



Revision history (moved here from the title block in V48, 13 Nov 2026, so that the title fits its box; wording unchanged):
Artix Frontend V47 – nets per pin map V6 (12 new top-chip inputs, top-chip SPI outputs, 2 clock lane swaps, VIN_5V on J1.32, spare J1.36), R7/R27 = JLC BOM parts
V46 J2 balls re-picked in lane order (12-layer lane trial: J2 58/58, J1 50/50)
V45 edge connectors: J1/J2 pin map V5, U1 re-pin, CLK25 N3->T14 (8L V3)
V44 cable pairs re-pinned near J3 (separate SYNC 12/14 to B9/A9, ASIC6_DATA6 P6->J6)
V43 J1/J2 remap (FPGA re-pin, LDO V3 pin order)
V42 J3 TX0 pins 3/5 swapped (TXPOLARITY=1)
V41 separate LVDS SYNC on HDMI 12/14 to D8/C8 (HDMI board V41 map) + V39 sourcing swaps (schematic for the re-route)
V47 comment 1: HDMI: 3/5 GT TX0 recording; 6/8 LVDS 32 MHz clock; 9/11 LVDS command+SYNC; 12/14 LVDS SYNC (ports 3-6) to B9/A9 (V44)
V47 comment 2: V47 schematic from V46 (12 Nov 2026, Howard: move forward, finish this up); for 8L V3 on 14 layers V4; not fabrication-ready
V48 (13 Nov 2026, finishing pass; Claude): title block text only. No symbol, wire, label or field changed; netlist identical to V47.

Not fabrication-ready. Revision history: note above this box
V50 = V49 with R20, R21, R49, R50 as a JLC-stocked 1 % part; nets unchanged; for 14 layers V7
HDMI: 3/5 GT TX0; 6/8 LVDS clock; 9/11 command+SYNC; 12/14 SYNC (ports 3-6)

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Title: Artix Frontend V50 – nets per pin map V6

Size: User	Date: 2026-11-13	**Rev: V50**
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